Curriculum Reinforcement Learning for Variable-Height Balancing of a 10-Actuated Wheeled Biped Robot

Van Thuong Nguyen
Independent Researcher
Email: vanthuong180702t@gmail.com

***Abstract*—This paper presents a curriculum reinforcement learning framework for training a 10-actuated wheeled biped robot to balance at variable commanded torso heights. Each leg has five actuated joints: hip roll, hip yaw, hip pitch, knee, and wheel. The control policy is trained in MuJoCo MJX with JAX-accelerated vectorized simulation using proximal policy optimization (PPO). The policy receives a 42-dimensional observation that includes proprioceptive state, a normalized height command, normalized current torso height, and a yaw-error signal, and outputs joint-level targets that are converted to actuator torques through a PID low-level controller. To stabilize learning across a wide range of body heights, we employ a within-stage height curriculum that progressively expands the commanded height range from near the nominal standing posture toward deep crouching configurations, gated by a reward-threshold criterion evaluated on a dedicated greedy evaluation pass approximately every 10^7 environment steps. A key component of the reward design is a height-consistent natural pose term that maps commanded height to target hip-pitch and knee angles via piecewise-linear interpolation, preventing degenerate postures that satisfy height objectives through unnatural joint configurations. We additionally implement a classical LQR baseline built on a linearized two-wheeled inverted pendulum (TWIP) model with height-dependent IK, providing a fair reference for the near-nominal operating range. The framework also includes a multi-stage macro-curriculum structure (stages: `balance`, `stand_up`, `balance_robust`) for future extension, though the present work focuses on the variable-height balancing stage. We describe the full system design, reward formulation, curriculum strategy, LQR baseline, and evaluation protocol, and present the framework as a foundation for further training and experimental validation.**

***Index Terms*—Wheeled biped robot, reinforcement learning, curriculum learning, variable-height balancing, PID control, LQR baseline, MuJoCo MJX.**

I. Introduction

Wheeled biped robots combine the efficiency of wheel-based locomotion with the adaptability of legged morphology. Compared with conventional legged systems, wheeled bipeds can achieve lower-cost motion on flat terrain while exploiting articulated legs to modify body posture, regulate center-of-mass height, and improve disturbance rejection. However, these benefits come at the cost of more challenging control: balancing at different commanded heights requires managing the nonlinear coupling between wheel dynamics, torso attitude, and multi-joint leg motions.

Classical control approaches [11] can stabilize wheeled bipeds near a nominal operating point, but become increasingly inaccurate as the robot moves away from the linearization height. Recent progress in reinforcement learning (RL) has shown that robust control policies can be learned directly in simulation for complex robotic systems [14], [15], [17]. Nevertheless, training a single policy to balance across a wide range of body heights is difficult due to poor sample efficiency and the risk of reward exploitation. In particular, simple height-tracking rewards may allow policies to satisfy torso height objectives through unnatural or unstable joint configurations rather than through physically meaningful postures.

To address these challenges, we propose a curriculum reinforcement learning framework for a 10-actuated wheeled biped robot. The method explicitly conditions the policy on a target torso height and uses a within-stage height curriculum to progressively expand the difficulty from near-nominal standing to deep crouching. A height-consistent posture reward regularizes the joint configuration as a function of the commanded height, improving training stability and posture quality. We also implement a classical LQR baseline that operates on the same observation interface and PID low-level controller as the RL policy, providing an interpretable reference for the regime where linear assumptions hold.

The main contributions of this work are:

- A height-conditioned RL framework for variable-height balancing of a 10-actuated wheeled biped, with PID low-level control bridging policy outputs to actuator torques.
- A within-stage height curriculum that expands the commanded height range based on a greedy evaluation pass gated at a reward-threshold criterion, stabilizing learning across the full feasible height range.
- A height-consistent natural pose reward that ties hip-pitch and knee joint targets to the commanded torso height via piecewise-linear interpolation, preventing degenerate balancing strategies.
- A classical LQR baseline built on a linearized TWIP model with height-dependent inverse kinematics, serving as an interpretable comparison for the near-nominal height regime.
- A quantitative evaluation benchmark with ten scenario groups and ten per-scenario metrics, enabling rigorous comparison between the RL policy and the LQR baseline.

The framework additionally includes a three-stage macro-curriculum: `balance`, `stand_up`, and `balance_robust`. It also includes a quantitative offline evaluation benchmark suite with ten scenario groups and ten per-scenario metrics. These extensions are described in this paper, though their full experimental results remain as future work.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the robot platform and simulation setup. Section IV formulates the learning problem. Section V describes the reward design. Section VI presents the curriculum strategy. Section VII describes the classical LQR baseline. Section VIII details the training pipeline and evaluation protocol. Section IX is reserved for experimental results. Section X discusses limitations and future work. Section XI concludes the paper.

II. Related Work

A. Classical and Model-Based Control for Wheeled Bipeds

Wheeled bipedal robots combine the energy efficiency of wheel-based locomotion with the postural adaptability of articulated legs. Early and influential work by Klemm *et al.* [10] introduced Ascento, a two-wheeled jumping robot demonstrating active balance through LQR-based stabilization around an inverted pendulum model. Subsequent work [11] extended this to an LQR-assisted whole-body controller with kinematic loops. Hsu *et al.* [12] applied fuzzy logic control to wheeled biped balance regulation. Li *et al.* [13] investigated dynamic motion control for wheel-biped transformable robots. The DIABLO platform [24] demonstrated six-DOF direct-drive wheeled biped control using cascaded LQR. While these model-based approaches provide stability guarantees near an operating point, they require accurate dynamics models and degrade when the robot deviates significantly from the linearization configuration—a limitation that becomes severe under variable-height commands.

B. Deep Reinforcement Learning for Legged Locomotion

Deep RL has produced robust locomotion policies for complex legged systems. Hwangbo *et al.* [17] introduced a learned actuator network that accurately models motor dynamics, enabling sim-to-real transfer of agile skills on ANYmal. Lee *et al.* [15] demonstrated blind proprioceptive locomotion over challenging outdoor terrain using a teacher-student distillation pipeline. Kumar *et al.* [16] proposed Rapid Motor Adaptation (RMA), a two-stage approach that trains an adaptation module online to handle unseen dynamics such as varying payload and slippery floors. Rudin *et al.* [14] showed that massively parallel GPU-based simulation (4,096 environments in Isaac Gym) can train walking policies in minutes using a terrain curriculum closely related to ours. Miki *et al.* [18] further demonstrated perception-aware locomotion over kilometer-scale outdoor terrain using attention-based proprioceptive encoders. These works establish that RL can scale to high-dimensional contact-rich control problems; our work adapts this paradigm to the specific challenge of variable-height wheeled-biped balancing.

C. Reinforcement Learning for Wheeled-Legged Robots

More recently, RL has been applied to wheeled-legged hybrid platforms. Lee *et al.* [19] presented a full autonomy stack for ANYmal-on-wheels, achieving 20 km/h driving with 83% reduced cost-of-transport. Vollenweider *et al.* [20] trained diverse locomotion skills for wheeled-legged robots through adversarial motion priors. Chamorro *et al.* [21] used asymmetric actor-critic RL with privileged terrain information for blind stair climbing on Ascento, successfully transferring to hardware for 15 cm step negotiation. Zhu *et al.* [22] introduced Whleaper, a 10-DOF wheeled biped most similar to our platform, demonstrating both LQR and PPO control for multimodal locomotion. Zhu and Hayashibe [23] applied hierarchical RL for path tracking on a wheeled biped with a sim-to-real transfer pipeline. In contrast to these platforms—which either fix body height or focus on forward locomotion—our work specifically addresses *variable-height stationary balance*, requiring the policy to operate stably across a 30 cm height range while maintaining postural integrity.

D. Curriculum Learning for Robot Control

Curriculum learning, introduced by Bengio *et al.* [9], organizes training from easy to hard examples to improve convergence and generalization. In robotics, Rudin *et al.* [14] demonstrated a terrain difficulty curriculum that substantially improves traversal capability. The Automatic Curriculum Learning (ACL) survey by Portelas *et al.* [25] systematizes curriculum strategies by learning progress, distinguishing goal-parameterized curricula from task-space curricula. Teacher algorithms such as ALP-GMM [26] generate parameterized task distributions that maximize learning progress in continuous task spaces. POET [27] co-evolves environments and agents to produce increasingly complex curricula. Our within-stage height curriculum is a monotone threshold-gated curriculum: the commanded height range expands only when a greedy evaluation pass confirms the policy has mastered the current range, directly addressing the instability that arises when a single policy must balance across configurations with qualitatively different inverted-pendulum dynamics.

E. Domain Randomization and Sim-to-Real Transfer

Sim-to-real transfer for locomotion policies relies on domain randomization to cover the reality gap. Tobin *et al.* [8] established domain randomization as a viable transfer strategy. Peng *et al.* [28] demonstrated dynamics randomization for robust quadruped locomotion on physical hardware. Tan *et al.* [29] showed that randomizing actuator parameters and adding observation noise enables Minitaur to transfer sim-trained trotting gaits to the real robot without fine-tuning. Our framework adopts these established protocols [8], [28], [29]—randomizing friction, mass properties, joint damping, and adding IMU/encoder noise and optional action delay—to prepare the trained policy for future hardware deployment.

III. Robot Platform and Simulation Setup

A. Robot Morphology

The robot considered in this work is a 10-actuated wheeled biped with five actuated joints on each leg: hip roll, hip yaw, hip pitch, knee, and wheel. The platform has a total mass of approximately $8.1\,\mathrm{kg}$, thigh length of $0.26\,\mathrm{m}$, shin length of $0.28\,\mathrm{m}$, wheel radius of $0.06\,\mathrm{m}$, and hip width of $0.23\,\mathrm{m}$. The sensing system includes one IMU and ten joint encoders. A physical robot with the same kinematic structure exists and is intended for future sim-to-real transfer, though hardware experiments are not part of the present work.

Let the joint vector be defined as

$$q_j = \begin{bmatrix} q^L_{\mathrm{hr}} & q^L_{\mathrm{hy}} & q^L_{\mathrm{hp}} & q^L_{\mathrm{kn}} & q^L_{\mathrm{wh}} & q^R_{\mathrm{hr}} & q^R_{\mathrm{hy}} & q^R_{\mathrm{hp}} & q^R_{\mathrm{kn}} & q^R_{\mathrm{wh}} \end{bmatrix}^\top, \quad (1)$$

where the superscripts L and R denote the left and right legs, respectively.

Fig. 1 illustrates the mechanical structure and joint layout. The articulated hip and knee joints enable variable-height posture regulation, while the wheel joints provide balancing capability through active torque generation.

B. Simulation Environment

All policies are trained in MuJoCo MJX [2], [3] with JAX-based [4] vectorized simulation. The physics simulation runs at $500\,\mathrm{Hz}$ with timestep

$$\Delta t_{\mathrm{sim}} = 0.002\,\mathrm{s}, \quad (2)$$

while the policy is queried at $50\,\mathrm{Hz}$,

$$\Delta t_{\pi} = 0.02\,\mathrm{s}. \quad (3)$$

Each policy action is therefore held constant for ten simulation substeps. Training is parallelized across 4096 environments on GPU [14].

Domain randomization [8] is applied during training to improve robustness. The base `balance` stage uses moderate perturbations to first learn clean stationary balance: body masses are perturbed by $\pm 10\%$, ground friction by $\pm 30\%$, and joint damping by $\pm 20\%$ relative to their nominal values. The later `balance_robust` stage widens these ranges to $\pm 15\%$, $\pm 40\%$, and $\pm 30\%$, respectively. Randomization is applied per episode at reset time. In the base `balance` stage, no external push disturbances are applied ($F_{\mathrm{push}} = 0$); random joint-position noise of $\pm 0.10\,\mathrm{rad}$ is added at reset to expose the policy to varied initial postures. Push disturbances ($40\,\mathrm{N}$ applied over 8 control steps, $\approx 0.16\,\mathrm{s}$) are introduced only in the `balance_robust` fine-tuning stage.

Table I summarizes the randomized parameters and their stage-specific ranges, following established sim-to-real protocols [8], [28]. Each parameter is sampled independently per episode from a uniform distribution over the stated range, with the nominal value at the midpoint unless noted.

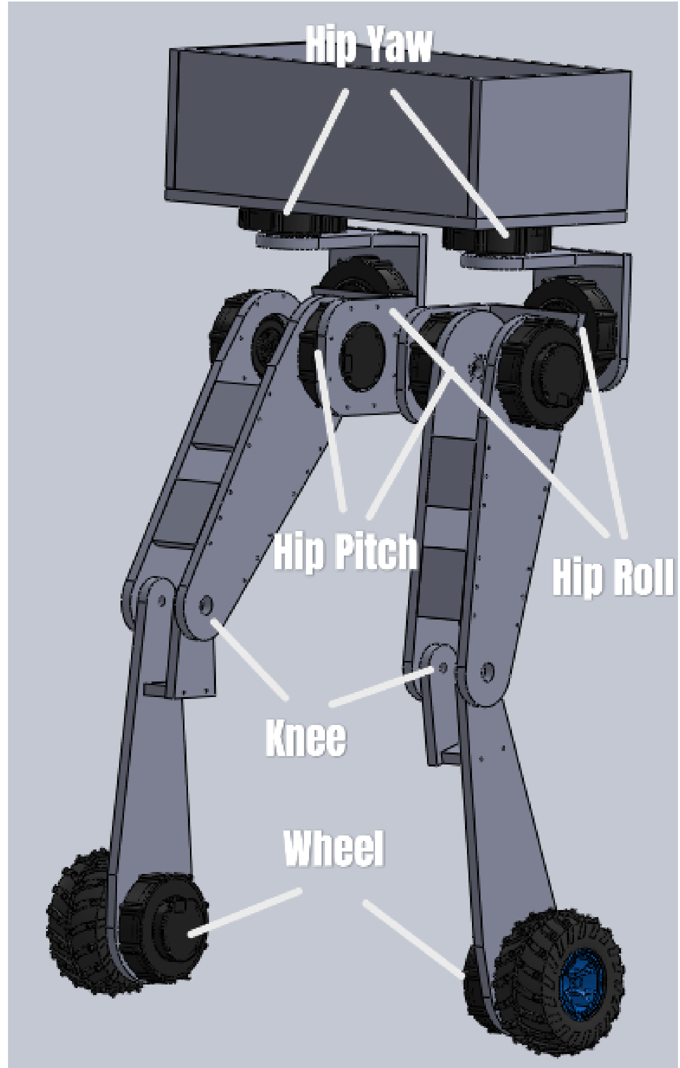

Fig. 1. Mechanical structure and joint layout of the wheeled biped robot. Each leg has five actuated joints: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). The left leg is annotated; the right leg is mirror-symmetric. The hip and knee joints enable variable-height posture regulation; the wheel joints provide balancing torque.

C. Low-Level PID Control

Rather than mapping policy outputs directly to actuator torques, we employ a PID low-level controller that converts normalized policy targets into torque commands. This decouples the policy from raw torque generation and simplifies the learning problem.

The policy outputs $a_t \in [-1, 1]^{10}$, which are interpreted differently depending on the joint type:

- **Leg joints** (hip roll, hip yaw, hip pitch, knee): the policy output is interpreted as a position target. A PD+I controller then computes the torque:

$$\tau_j = K_p^j (q_j^* - q_j) + K_d^j (0 - \dot{q}_j) + K_i^j \int (q_j^* - q_j)\, dt, \quad (4)$$

  where q_j^* is the position target, q_j and $\dot{q}_j$ are the measured joint position and velocity, and K_p^j, K_d^j, K_i^j are per-joint

TABLE I

DOMAIN RANDOMIZATION PARAMETERS APPLIED DURING TRAINING. ALL PARAMETERS ARE SAMPLED UNIFORMLY PER EPISODE. SENSOR NOISE IS SAMPLED INDEPENDENTLY PER STEP.

Parameter	Range	Unit
Dynamics		
Friction coefficient	Balance: $[0.7,\ 1.3]\times$; robust: $[0.6,\ 1.4]\times$	—
Total body mass	Balance: $[0.9,\ 1.1]\times$; robust: $[0.85,\ 1.15]\times$	—
Joint damping	Balance: $[0.8,\ 1.2]\times$; robust: $[0.7,\ 1.3]\times$	—
Sensors & Actuation		
IMU angular velocity noise	$\mathcal{N}(0,\ 0.05^2)$	rad/s
Gravity-vector noise	$\mathcal{N}(0,\ 0.02^2)$	normalized
Joint position noise	$\mathcal{N}(0,\ 0.005^2)$	rad
Joint velocity noise	$\mathcal{N}(0,\ 0.01^2)$	rad/s
Linear velocity mode	clean; optional noisy ($\sigma = 0.2$)	m/s
Action delay	0 default; optional 1 step	—
Initial Conditions		
Initial joint offset	$[-0.10,\ 0.10]$	rad

TABLE II

PER-JOINT PID GAINS. JOINT ORDER: HIP ROLL (HR), HIP YAW (HY), HIP PITCH (HP), KNEE (KN), WHEEL (WH). GAINS ARE IDENTICAL FOR LEFT AND RIGHT LEGS. THE WHEEL K_d IS MASKED TO ZERO INTERNALLY REGARDLESS OF THE TABLE VALUE.

Joint	K_p	K_i	K_d
HR	55.0	0.8	3.0
HY	40.0	0.4	2.0
HP	70.0	1.0	4.0
KN	70.0	1.0	4.0
WH	4.0	0.1	(0.0)

gains. The integral term includes anti-windup clamping ($\pm 0.4\,\mathrm{rad{\cdot}s}$).

- **Wheel joints**: the policy output is interpreted as a velocity target. A PI controller computes the torque:

$$\tau_w = K_p^w(\dot{q}_w^* - \dot{q}_w) + K_i^w \int (\dot{q}_w^* - \dot{q}_w)\, dt, \quad (5)$$

where $\dot{q}_w^*$ is the velocity target. The derivative gain is masked to zero ($K_d^w = 0$) because wheel velocity-error derivative is the joint acceleration rather than joint velocity, making standard K_d semantics ill-defined for velocity-control joints.

An action-smoothing filter with coefficient $\alpha = 0.4$ blends the current and previous policy output before passing to the PID controller, reducing command chattering. Optionally, an action delay of n_{delay} control steps can model CAN-bus latency during sim-to-real preparation (default: $n_{\text{delay}} = 0$ for simulation training). All torque outputs are clipped to the actuator limits. The per-joint gains are given in Table II.

IV. PROBLEM FORMULATION

A. State, Observation, and Action

At time step t, the robot state is $x_t = \{q_t, \dot{q}_t\}$, where q_t and $\dot{q}_t$ are the generalized coordinates and velocities.

The policy receives a **42-dimensional** observation vector

$$o_t = \left[g_t^b,\ v_t^b,\ \omega_t^b,\ q_{j,t},\ \dot{q}_{j,t},\ a_{t-1},\ \bar{h}_t^{cmd},\ \bar{z}_t,\ e_{\psi,t}\right] \in \mathbb{R}^{42}, \quad (6)$$

where:

- $g_t^b \in \mathbb{R}^3$: gravity vector expressed in the body frame (from IMU accelerometer);
- $v_t^b \in \mathbb{R}^3$: body-frame linear velocity;
- $\omega_t^b \in \mathbb{R}^3$: body-frame angular velocity;
- $q_{j,t} \in \mathbb{R}^{10}$: joint positions;
- $\dot{q}_{j,t} \in \mathbb{R}^{10}$: joint velocities;
- $a_{t-1} \in \mathbb{R}^{10}$: previous smoothed policy action;
- $\bar{h}_t^{cmd} \in \mathbb{R}$: normalized height command;
- $\bar{z}_t \in \mathbb{R}$: normalized current torso height;
- $e_{\psi,t} \in \mathbb{R}$: yaw error (heading drift from episode start).

The first six entries in Eq. (6) form the base 39-dimensional observation inherited from the base environment. The three task-specific extensions are:

Normalized height command.

$$\bar{h}_t^{cmd} = \frac{h_t^{cmd} - h_{\min}}{h_{\max} - h_{\min}}, \quad (7)$$

where $h_{\min} = 0.40\,\mathrm{m}$ and $h_{\max} = 0.70\,\mathrm{m}$. The normalization always uses the full task range, so the value is in $[0, 1]$ regardless of the current curriculum level.

Normalized current torso height.

$$\bar{z}_t = \frac{z_t - h_{\min}}{h_{\max} - h_{\min}}, \quad (8)$$

where z_t is the current torso height. This term exposes instantaneous height-tracking error directly to the memoryless policy.

Yaw error.

$$e_{\psi,t} = \mathrm{wrap}(\psi_t - \psi_0)\,, \quad (9)$$

where ψ_t is the current torso yaw extracted from the base quaternion via Euler decomposition, ψ_0 is the yaw at episode reset, and $\mathrm{wrap}(\cdot)$ maps to $(-\pi, \pi]$. This term is necessary because the gravity-body-frame component g_t^b is yaw-invariant: a memoryless MLP cannot perceive accumulated heading drift from g^b and ω^b alone.

When linear velocity is unavailable (e.g., in sim-to-real preparation without a reliable velocity estimator), v_t^b can be removed, reducing the observation to 39 dimensions. All results in this paper use simulator-exact linear velocity, which corresponds to the upper bound on velocity-estimate quality.

Observations are normalized online using Welford's running mean and variance estimator [7], updated during training rollouts and held fixed during evaluation.

The policy outputs a normalized action $a_t = \pi_\theta(o_t) \in [-1, 1]^{10}$, converted to actuator torques through the PID controller described in Section III-C.

B. Learning Objective

We seek a policy π_θ that maximizes the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], \quad (10)$$

TABLE III

REWARD COMPONENTS AND WEIGHTS FOR THE VARIABLE-HEIGHT BALANCE TASK. WEIGHTS FOR THE BALANCE_ROBUST FINE-TUNING STAGE DIFFER: $w_{\text{still}} = 0$, $w_{\text{wh}} = 0$, $w_{\text{nat}} = 1.5$, $w_{\text{alive}} = 0.5$, $w_{\text{drift}} = 1.0$, AND $w_{\Delta a} = -0.005$, TO ACCOMMODATE WHEEL-BASED PUSH RECOVERY MOTION.

Component	Type	Weight
Height tracking (r_{height})	Positive	2.5
Body level (r_{level})	Positive	1.5
Position drift (r_{drift})	Positive	2.5
Orientation (r_{orient})	Positive	1.0
Symmetry (r_{sym})	Positive	1.0
Heading (r_{heading})	Positive	0.5
Legs forward (r_{fwd})	Positive	0.5
Legs vertical (r_{vert})	Positive	0.5
No motion (r_{still})	Positive	1.0
Yaw suppression (r_{yaw})	Positive	0.5
Natural pose (r_{nat})	Positive	0.4
Alive bonus (r_{alive})	Positive	0.3
Action rate ($p_{\Delta a}$)	Penalty	-0.08
Wheel velocity (p_{wh})	Penalty	-0.006
Joint torque (p_{τ})	Penalty	-0.0008
Joint velocity ($p_{\dot{q}}$)	Penalty	-0.001

where $\gamma = 0.99$ and r_t is the reward at step t.

Training is performed using PPO [1] with clipped surrogate objective

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t,\ \text{clip}\big(\rho_t(\theta), 1-\epsilon, 1+\epsilon\big)\hat{A}_t\Big)\Big], \tag{11}$$

where $\rho_t(\theta) = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ and $\epsilon = 0.20$. The full objective adds a clipped value loss (coefficient 1.0) and entropy regularization (coefficient 0.005 for the `balance` stage). Advantages $\hat{A}_t$ are computed using Generalized Advantage Estimation (GAE) [6] with $\lambda = 0.95$.

V. REWARD DESIGN

A. Reward Structure

The total reward at each step is a weighted sum:

$$r_t = \sum_i w_i\, r_t^{(i)}. \tag{12}$$

The complete set of reward terms and weights for the `balance` task are given in Table III.

B. Primary Reward Terms

1) Body-Level Reward: To encourage the torso to remain parallel to the ground:

$$r_{\text{level}} = \exp\left(-\frac{\phi_t^2}{\sigma_r^2}\right)\exp\left(-\frac{\theta_t^2}{\sigma_p^2}\right), \tag{13}$$

where ϕ_t and θ_t are the torso roll and pitch angles, and $\sigma_r = \sigma_p = 0.15\,\text{rad}$. The threshold $\sigma = 0.15\,\text{rad}$ ($\approx 8.6°$) corresponds to the boundary of stable balance for a wheeled inverted pendulum at the nominal height: beyond this lean angle the TWIP linearization becomes inaccurate and recovery is uncertain. This choice ensures the reward remains near 1.0 during upright stance and falls sharply only when tilt enters the unstable regime, providing a tight but achievable target for the policy.

2) Height Tracking Reward:

$$r_{\text{height}} = \exp\left(-\frac{(h_t - h_t^{cmd})^2}{\sigma_h^2}\right), \tag{14}$$

with $\sigma_h = 0.25\,\text{m}$. This bandwidth was selected to provide a non-vanishing gradient signal across the full commanded height range $[0.40, 0.70]\,\text{m}$: at the midpoint of a 0.15 m tracking error, the reward retains $\exp(-0.15^2/0.25^2) \approx 0.83$ of its maximum value, ensuring a smooth gradient that guides learning even during early training when height tracking is poor. A tighter σ would produce near-zero gradients for heights far from the target and destabilize early-stage curriculum learning.

3) Heading and Position Regularization: To keep the robot near its initial yaw ψ_0 and anchor position $p_{xy,0}$:

$$r_{\text{heading}} = \exp\left(-\frac{\text{wrap}(\psi_t - \psi_0)^2}{\sigma_\psi^2}\right), \tag{15}$$

$$r_{\text{drift}} = \exp\left(-\frac{\|p_{xy,t} - p_{xy,0}\|_2^2}{\sigma_d^2}\right), \tag{16}$$

with $\sigma_\psi = 0.5\,\text{rad}$. For the `balance` stage we set $\sigma_d = 0.25\,\text{m}$ to penalize rolling drift early; recovery-oriented stages may use a wider drift scale.

4) Orientation and Yaw Stability: Linear orientation stability (avoids saturation at high angular rates):

$$r_{\text{orient}} = \text{clip}\big(1 - 0.15\,\|\omega_t^b\|_2,\ 0,\ 1\big), \tag{17}$$

and a dedicated yaw suppression term:

$$r_{\text{yaw}} = \text{clip}\big(1 - 0.5\,|\omega_{z,t}^b|,\ 0,\ 1\big). \tag{18}$$

5) Leg Symmetry:

$$r_{\text{sym}} = \exp\left(-\frac{\left\|q_{\text{leg},t}^L - q_{\text{leg},t}^R\right\|_2^2}{\sigma_{\text{sym}}^2}\right), \tag{19}$$

where $q_{\text{leg}}^{L/R}$ includes hip roll, hip yaw, hip pitch, and knee for each leg, and $\sigma_{\text{sym}} = 0.5$.

6) Leg Alignment Rewards: Two terms encourage well-aligned leg orientation:

Legs forward (r_{fwd}): rewards hip yaw close to zero (feet pointing forward):

$$r_{\text{fwd}} = \exp\left(-\frac{(q_{\text{hy}}^L)^2}{\sigma_\ell^2}\right)\exp\left(-\frac{(q_{\text{hy}}^R)^2}{\sigma_\ell^2}\right). \tag{20}$$

Legs vertical (r_{vert}): rewards hip roll close to zero (legs perpendicular to the ground plane):

$$r_{\text{vert}} = \exp\left(-\frac{(q_{\text{hr}}^L)^2}{\sigma_\ell^2}\right)\exp\left(-\frac{(q_{\text{hr}}^R)^2}{\sigma_\ell^2}\right), \tag{21}$$

where $\sigma_\ell = 0.15\,\text{rad}$ for both terms.

7) No-Motion Reward: To encourage stationary balance:

$$r_{\text{still}} = \exp\left(-\frac{\|v_t^b\|_2^2}{\sigma_v^2}\right), \tag{22}$$

The velocity scale is $\sigma_v = 0.2\,\text{m/s}$. This term is set to zero in the `balance_robust` stage, where wheel motion is needed for push recovery.

C. Height-Consistent Natural Pose Reward

A key design of our method is a posture regularization term that enforces consistency between the commanded torso height and the desired leg configuration. Let the desired hip-pitch and knee angles $q^*_{\mathrm{hp}}(h^{cmd})$ and $q^*_{\mathrm{kn}}(h^{cmd})$ be computed by piecewise-linear interpolation.

The commanded height range used by the balance-family policies is $h^{cmd} \in [0.40, 0.70]\,\mathrm{m}$. The interpolation map below uses slightly wider kinematic anchors to cover reset perturbations and mechanical limits; the anchor values $0.38\,\mathrm{m}$ and $0.72\,\mathrm{m}$ are not additional sampled command levels. Let h denote the height supplied to this interpolation map ($h = h^{cmd}$ during normal command tracking).

The kinematic keypoints, derived from forward kinematics of the two-link leg model, are:

$$
\begin{aligned}
h = 0.72\,\mathrm{m}:&\quad q^*_{\mathrm{hp}} = 0.0, \quad q^*_{\mathrm{kn}} = 0.0\\
h = 0.71\,\mathrm{m}:&\quad q^*_{\mathrm{hp}} = 0.3, \quad q^*_{\mathrm{kn}} = 0.5\\
h = 0.38\,\mathrm{m}:&\quad q^*_{\mathrm{hp}} = 1.5, \quad q^*_{\mathrm{kn}} = 2.5
\end{aligned}
$$

For the high-standing anchor segment $h \in [0.71, 0.72]\,\mathrm{m}$:

$$t_h = \frac{h - 0.71}{0.01}, \quad q^*_{\mathrm{hp}} = 0.3(1 - t_h), \quad q^*_{\mathrm{kn}} = 0.5(1 - t_h). \tag{23}$$

For the lower-height anchor segment $h \in [0.38, 0.71]\,\mathrm{m}$ (which contains the full commanded training range):

$$t_l = \frac{h - 0.38}{0.33}, \quad q^*_{\mathrm{hp}} = 1.5 - 1.2\,t_l, \quad q^*_{\mathrm{kn}} = 2.5 - 2.0\,t_l. \tag{24}$$

The posture error is

$$e_{\mathrm{pose}} = \left((q^L_{\mathrm{hp}} - q^*_{\mathrm{hp}})^2 + (q^R_{\mathrm{hp}} - q^*_{\mathrm{hp}})^2 + (q^L_{\mathrm{kn}} - q^*_{\mathrm{kn}})^2 + (q^R_{\mathrm{kn}} - q^*_{\mathrm{kn}})^2\right)^{1/2}. \tag{25}$$

The natural pose reward is

$$r_{\mathrm{nat}} = \exp\left(-\frac{e^2_{\mathrm{pose}}}{\sigma^2_{nat}}\right) r_{\mathrm{knee}}, \tag{26}$$

where $\sigma_{\mathrm{nat}} = 0.5\,\mathrm{rad}$ and r_{knee} penalizes knee hyperextension:

$$r_{\mathrm{knee}} = \exp\left(-\frac{\min(q^L_{\mathrm{kn}}, 0)^2}{\sigma_k^2}\right) \exp\left(-\frac{\min(q^R_{\mathrm{kn}}, 0)^2}{\sigma_k^2}\right), \tag{27}$$

with $\sigma_k = 0.15\,\mathrm{rad}$.

This reward prevents the policy from exploiting unnatural joint configurations (e.g., extreme hip abduction with nearly straight knees) to satisfy a height target without a kinematically meaningful posture.

Design rationale. Without this term, a height-tracking reward alone permits degenerate solutions: the robot can achieve the target CoM height by bending one leg more than the other (asymmetric collapse), hyperextending the knee, or adopting an extreme hip-pitch that satisfies the height constraint but produces an unstable inverted-pendulum configuration. The piecewise-linear interpolation maps each commanded height to a physically meaningful (hip-pitch, knee) pair derived from the two-link leg geometry, effectively constraining the null space of the height objective to stable, symmetric postures. The soft exponential penalty ($\sigma_{\mathrm{nat}} = 0.5\,\mathrm{rad}$) allows exploration during early training while strongly penalizing configurations that deviate more than $\approx 0.5\,\mathrm{rad}$ from the target posture.

D. Effort and Smoothness Penalties

We regularize torque magnitude, joint velocity, action variation, and wheel rotation:

$$p_\tau = \sum_{j=1}^{10} \tau^2_{t,j}, \quad p_{\dot q} = \sum_{j=1}^{10} \dot q^2_{j,t}, \quad p_{\Delta a} = \sum_{j=1}^{10} (a_{t,j} - a_{t-1,j})^2, \tag{28}$$

$$p_{\mathrm{wh}} = \dot q^{L\,2}_{\mathrm{wh},t} + \dot q^{R\,2}_{\mathrm{wh},t}. \tag{29}$$

The wheel velocity penalty and r_{still} together enforce stationary balance in the `balance` stage; both are set to zero in `balance_robust`.

E. Termination Conditions

An episode terminates early if the torso tilt exceeds $0.8\,\mathrm{rad}$ ($\approx 46°$) or the torso height drops below $0.3\,\mathrm{m}$. Otherwise, episodes run for 1000 steps ($\approx 20\,\mathrm{s}$). A survival bonus $r_{\mathrm{alive}} \in \{0, 1\}$ is given at each non-terminal step, providing a constant per-step incentive to remain standing.

VI. Curriculum Learning Strategy

A. Within-Stage Height Curriculum

Rather than sampling the commanded height from the full feasible range from the beginning, we employ a within-stage curriculum [9] that progressively expands the lower bound of the height command distribution.

The height lower bound is decreased by a fixed step δ_h only when all curriculum gate conditions pass:

$$h^{(k+1)}_{\min} = \max\big(h^{(k)}_{\min} - \delta_h,\ h^{\mathrm{final}}_{\min}\big), \quad \text{if } \mathcal{G}_{\mathrm{curr}} = 1. \tag{30}$$

Advancement signal. Every 10^7 environment steps, corresponding to 77 PPO updates under the 4096-environment, 32-step rollout configuration, the current policy is evaluated greedily (no exploration noise) using a JIT-compiled fixed-horizon rollout with 256 parallel evaluation environments. Each environment contributes one complete 1000-step episode, and the observation normalizer is held fixed. The per-step return is

$$\bar r_{\mathrm{eval}} = \frac{R_{\mathrm{mean}}}{T}, \tag{31}$$

where R_{mean} is the mean episode return and $T = 1000$ is the episode length. Advancement fires only when all gates pass:

$$
\begin{aligned}
\bar r_{\mathrm{train}} &\geq 7.0,\\
\bar r_{\mathrm{eval}} &\geq 0.70\, r_{\max},\\
s_{\mathrm{eval}} &\geq 0.99,\\
f_{\mathrm{eval}} &\leq 0.01.
\end{aligned} \tag{32}
$$

Here $\mathcal{G}_{\mathrm{curr}} = 1$ denotes that the conjunction in Eq. (32) holds. where s_{eval} and f_{eval} are the evaluation success and fall rates. The sum of positive `balance` reward weights is $r_{\max} = 12.2$, so the per-step evaluation threshold is $r_{\mathrm{thr}} = 0.70 r_{\max} =$

8.54 reward/step. Using a greedy evaluation pass rather than the noisy per-update training reward produces a more reliable signal for curriculum decisions.

Curriculum parameters. Initial lower bound: $h_{\min}^{\text{init}} = 0.69\,\text{m}$; final lower bound: $h_{\min}^{\text{final}} = 0.40\,\text{m}$; $N_c = 29$ discrete levels; $\delta_h \approx 0.01\,\text{m}$; threshold ratio 0.70 ($r_{\text{thr}} = 8.54$ reward/step), with the additional train-reward, success-rate, and fall-rate gates above. At each level k:

$$h_t^{cmd} \sim \mathcal{U}\big(h_{\min}^{(k)},\ h_{\max}\big), \tag{33}$$

where $h_{\max} = 0.70\,\text{m}$.

The 29 levels span three qualitative phases:

- **Phase A** (levels 1–5): $h_{\min} \in [0.65, 0.69]\,\text{m}$; narrow band near nominal; policy focuses on stable upright stance.
- **Phase B** (levels 6–20): $h_{\min} \in [0.50, 0.64]\,\text{m}$; widening toward moderate crouching.
- **Phase C** (levels 21–29): $h_{\min} \in [0.40, 0.49]\,\text{m}$; full operating range including deep crouching.

B. Multi-Stage Macro-Curriculum

A multi-stage curriculum manager chains training stages with warm-starting and performance-gated promotion and demotion. The three stages are:

1) **`balance`** (this paper): variable-height stationary balance with within-stage height curriculum as described above. No external push disturbances ($F_{\text{push}} = 0$). Macro-curriculum promotion threshold: 80% success rate with $w_{\text{eval}} = 7.0\,\text{reward/step}$.
2) **`stand_up`**: height transitions over the same commanded range $h^{cmd} \in [0.40, 0.70]\,\text{m}$ and recovery from fallen or non-nominal initial poses. This stage keeps the same 42-dimensional observation and PID action interface as `balance`, enabling direct checkpoint transfer without changing the policy architecture. Initialized (warm-started) from the `balance` checkpoint. Promotion threshold: 80% success rate with $w_{\text{eval}} = 5.0\,\text{reward/step}$.
3) **`balance_robust`**: fine-tuning under periodic random push disturbances ($F_{\text{push}} = 40\,\text{N}$, applied over 8 control steps, every 200 steps $\approx 4\,\text{s}$). It preserves the same observation and low-level action interface and is initialized from the `stand_up` checkpoint. Reward weights modified: $w_{\text{still}} = w_{\text{wh}} = 0$; w_{nat} increased to 1.5; $w_{\Delta a}$ reduced to -0.005. Promotion threshold: 80% success rate with $w_{\text{eval}} = 6.0\,\text{reward/step}$.

Demotion fires when success rate falls below 30% within a window of 100 evaluation episodes. Stages `stand_up` and `balance_robust` have corresponding environment implementations and training configurations but have not yet been trained; their results are not reported here.

VII. Classical LQR Baseline

A. Motivation

Classical control methods provide an interpretable reference for evaluating RL policies. We implement an LQR baseline that operates on the same 42-dimensional observation interface and shares the same PID low-level controller as the RL policy, enabling a controlled comparison without confounding actuator-interface differences.

B. Linearized TWIP Model

The sagittal balance component is based on a linearized two-wheeled inverted pendulum (TWIP) model. The state and input are

$$x_{\text{lqr}} = \begin{bmatrix} \theta_{\text{pitch}} \\ \dot{\theta}_{\text{pitch}} \\ v_{\text{fwd}} \\ p_{\text{fwd}} \end{bmatrix}, \quad u = \bar{\omega}_{\text{wheel}}, \tag{34}$$

where θ_{pitch} is the forward lean angle, v_{fwd} is the forward body velocity, p_{fwd} is the forward position drift, and u is the average wheel velocity command.

The continuous-time linearization $\dot{x}_{\text{lqr}} = A\,x_{\text{lqr}} + B\,u$ uses robot constants: mass $m = 8.1\,\text{kg}$, nominal CoM height $l_c = 0.54\,\text{m}$ (updated per height command via IK), and wheel radius $r_w = 0.06\,\text{m}$.

The LQR gain is solved from the continuous algebraic Riccati equation (CARE) with state cost $Q = \text{diag}(10, 2, 3, 0.3)$ and input cost $R = 0.8$:

$$K_{\text{lqr}} = R^{-1}B^{\top}P, \quad A^{\top}P + PA - PBR^{-1}B^{\top}P + Q = 0. \tag{35}$$

C. Height-Dependent Inverse Kinematics

To extend the LQR to variable commanded heights, we perform an offline IK scan over hip-pitch angles from 0 to $1.35\,\text{rad}$, with the knee set to $q_{\text{kn}} = 2\,q_{\text{hp}}$ (symmetric two-link fold geometry). The resulting height–angle mapping is fit by a degree-2 polynomial, yielding target joint angles for any commanded height in $[0.40, 0.70]\,\text{m}$. The effective CoM height $l_c(h^{cmd})$ fed into the TWIP linearization is updated accordingly.

D. Multi-Axis Control

Lateral and yaw axes are controlled by independent PD terms:

$$u_{\text{lat}} = K_{p,\text{roll}} \cdot e_{\text{roll}} + K_{d,\text{roll}} \cdot \dot{e}_{\text{roll}}, \tag{36}$$

$$u_{\text{yaw}} = K_{p,\text{yaw}} \cdot e_{\psi} + K_{d,\text{yaw}} \cdot \omega_z, \tag{37}$$

with $K_{p,\text{roll}} = 0.4$, $K_{d,\text{roll}} = 0.08$, $K_{p,\text{yaw}} = 2.5$, $K_{d,\text{yaw}} = 0.2$. The final wheel command is the superposition of sagittal LQR, lateral PD, and yaw PD outputs, clipped to $\pm 20.0\,\text{rad/s}$.

E. Assumptions and Limitations

The LQR baseline has the following known limitations:

1) **Linear regime only.** The TWIP linearization is valid near the nominal standing height and small lean angles ($|\theta_{\text{pitch}}| \lesssim 10°$). At large lean deviations or very low commanded heights, accuracy degrades.
2) **No lean integral.** Without an integral term on lean error, constant disturbances (friction variation, slopes) produce nonzero steady-state lean.

3) **No large-disturbance recovery.** Under strong pushes the state leaves the linear regime; the LQR provides no recovery mechanism.
4) **Decoupled axes.** Sagittal, lateral, and yaw dynamics are treated independently, ignoring cross-coupling during aggressive maneuvers.

The LQR baseline is therefore a fair reference for the `nominal` and `narrow_height` scenarios but is expected to degrade at full height range, strong pushes, and low friction.

VIII. Training Pipeline and Evaluation Protocol

A. Network Architecture

Separate MLPs for policy and value function: hidden sizes $\{256, 256, 128\}$, ELU activation, orthogonal weight initialization, implemented in Flax [5]. The policy outputs the mean of a Gaussian action distribution with a learned log-standard-deviation initialized to $\log(0.2)$.

B. Training Configuration

PPO with 4096 parallel environments, rollout length 32 steps, 4 epochs per update, 4 minibatches, learning rate $3{\times}10^{-4}$ (Adam), gradient norm clipping 0.5, entropy coefficient 0.005, value-loss coefficient 1.0. To avoid rare destructive policy jumps during long resumed runs, an update is rejected if its approximate policy KL exceeds 0.03. Long training runs use a single NVIDIA Tesla V100-SXM2 GPU with 32 GB memory, an Intel Xeon Gold 6248 CPU at 2.50 GHz, and 94 GB of system RAM.

The complete set of hyperparameters is summarized in Table IV for reproducibility.

TABLE IV
PPO training hyperparameters for the `balance` stage.

Hyperparameter	Value
Environment	
Training GPU	NVIDIA Tesla V100-SXM2, 32 GB
CPU / system RAM	Intel Xeon Gold 6248, 2.50 GHz / 94 GB
Parallel environments	4,096
Rollout length	32 steps
Control frequency	50 Hz (20 ms/step)
Episode length	1,000 steps (20 s)
PPO Update	
Epochs per update	4
Minibatches per epoch	4
Minibatch size	32,768
PPO clip range (ϵ)	0.20
Approx. KL reject threshold	0.03
Optimization	
Optimizer	Adam
Learning rate	3×10^{-4}
Gradient norm clip	0.5
Value loss coefficient	1.0
Entropy coefficient	0.005
Advantage Estimation	
Discount γ	0.99
GAE λ	0.95
Network	
Hidden layers (actor)	$\{256, 256, 128\}$
Hidden layers (critic)	$\{256, 256, 128\}$
Activation	ELU
Weight init	Orthogonal
Initial log-std	$\log(0.2)$
Curriculum Evaluation	
Eval interval	every 10^7 env steps (≈ 77 updates)
Eval environments	256
Eval episodes	256

C. Research Evaluation Protocol

Research evaluation is performed offline using a deterministic single-environment MuJoCo rollout with per-step telemetry recording. This evaluation pipeline is strictly separate from the in-training curriculum advancement evaluation, which uses a vectorized greedy rollout over the current policy to gate height-range expansion.

Evaluation scenarios. Ten scenario groups are defined:

1) **Nominal**: flat floor, $h^{cmd} = 0.65\,\mathrm{m}$, standard domain randomization.
2) **Narrow height**: $h^{cmd} = 0.69\,\mathrm{m}$ (Phase A, near-nominal).
3) **Wide height**: $h^{cmd} = 0.60\,\mathrm{m}$ (Phase B, moderate crouching).
4) **Full range**: $h^{cmd} \sim \mathcal{U}[0.40, 0.70]$ (Phase C, full range).
5) **Push recovery**: single $50\,\mathrm{N}$ impulse at $t = 3\,\mathrm{s}$; measures recovery time and maximum recoverable disturbance via binary search (range $10\,\mathrm{N}$ to $300\,\mathrm{N}$, 8 iterations, 50% survival threshold).
6) **Friction low**: friction multiplier $0.6\times$.
7) **Friction high**: friction multiplier $1.4\times$.
8) **Sensor noise + delay**: IMU/encoder noise plus 1-step ($\approx 20\,\mathrm{ms}$) action delay.
9) **Push sweep**: parametric sweep over eight push magnitudes from $20\,\mathrm{N}$ to $200\,\mathrm{N}$ for degradation curves.
10) **Friction sweep**: parametric sweep over friction multipliers $\{0.3, 0.5, 0.7, 1.0, 1.3, 1.8\}$.

Evaluation metrics. Ten metrics are computed per scenario:

1) Survival time [s]
2) Pitch RMS [deg]
3) Roll RMS [deg]
4) Pitch-rate RMS [rad/s]
5) XY drift max [m]
6) Height RMSE [m]
7) Wheel speed RMS [rad/s]
8) Torque RMS [N·m]
9) Recovery time [s] (push_recovery only; NaN otherwise)
10) Max recoverable push [N] (push_recovery only)

Aggregate metrics: fall rate, survival rate.

Multi-seed evaluation protocol. Final RL results are reported as mean ± standard deviation over three independent training runs initialized with random seeds 42, 113, and 999. Each run is trained from scratch with a distinct random state; no parameters, replay data, or normalization statistics are shared between runs during training or evaluation. The offline evaluation pipeline is applied identically to all three trained

TABLE V
RL POLICY VS. LQR BASELINE COMPARISON (PLANNED). ALL RESULT CELLS TO BE FILLED AFTER TRAINING.

	Scenario	Surv. (%)	e_h (m)	Pitch (deg)	Torque (Nm)	Push (N)
RL	Nominal	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Narrow	[TODO]	[TODO]	[TODO]	[TODO]	—
	Full range	[TODO]	[TODO]	[TODO]	[TODO]	—
	Push recov.	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Friction low	[TODO]	[TODO]	[TODO]	[TODO]	—
LQR	Nominal	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Narrow	[TODO]	[TODO]	[TODO]	[TODO]	—
	Full range	[TODO]	[TODO]	[TODO]	[TODO]	—
	Push recov.	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Friction low	[TODO]	[TODO]	[TODO]	[TODO]	—

policies, and scalar metrics are aggregated post-hoc. The LQR baseline requires no training and is evaluated once under the same scenario suite.

IX. Experimental Results

Training has not yet been completed. This section will be populated with quantitative results after the `balance` *policy is fully trained and evaluated. All placeholders below indicate planned experiments.*

A. Learning Curves

[TODO-FIGURE: training curve and curriculum level progression]

B. Nominal Balance and Variable-Height Results

TABLE VI
VARIABLE-HEIGHT BALANCING: PERFORMANCE VS. COMMANDED HEIGHT. ALL VALUES PENDING TRAINING COMPLETION.

h^{cmd} (m)	Surv. (%)	e_h (m)	Pitch (deg)	Torque (Nm)
0.40	[TODO]	[TODO]	[TODO]	[TODO]
0.50	[TODO]	[TODO]	[TODO]	[TODO]
0.60	[TODO]	[TODO]	[TODO]	[TODO]
0.65	[TODO]	[TODO]	[TODO]	[TODO]
0.69	[TODO]	[TODO]	[TODO]	[TODO]

C. Push Robustness

[TODO-FIGURE: push_sweep degradation curve; TODO-RESULT: max recoverable push force]

D. Friction Robustness

[TODO-FIGURE: friction_sweep degradation curve]

E. Ablation: Natural Pose Reward

[TODO-ABLATION: natural pose reward (r_{nat}) ablation: weight=0.4 vs. weight=0]

F. Ablation: Observation Design

[TODO-ABLATION: velocity-estimate robustness (exact / noisy / absent) — only report if experiment is conducted]

X. Discussion and Limitations

A. Framework Design

The main design principle is to decompose the variable-height balancing problem into a progressive curriculum that stabilizes learning by starting near the nominal standing pose and expanding toward more challenging configurations. The height-consistent natural pose reward constrains the joint-space solution to physically meaningful postures at each target height.

The PID low-level controller decouples the RL policy from actuator-level dynamics and is shared with the LQR baseline, ensuring that observed performance differences reflect controller intelligence rather than interface differences.

The yaw-error observation term $e_{\psi,t}$ closes the heading loop that gravity-body-frame observations alone cannot. Without it, a memoryless MLP requires undesirable implicit integration of angular velocity over many steps to sense accumulated yaw drift.

B. Sim-to-Real Considerations

Although the present work is conducted entirely in simulation, the framework is designed with hardware transfer in mind.

Action representation. The policy outputs PD position targets rather than raw torques. This action representation has been shown to transfer reliably across the sim-to-real gap for legged locomotion [14], [17], as it implicitly low-pass filters the control signal and decouples the policy from precise actuator gain tuning.

Domain randomization coverage. The randomization ranges in Table I were chosen to separate nominal balance learning from later robustness fine-tuning. The `balance` stage uses moderate perturbations to avoid learning spurious wheel-spin or drift strategies before a clean stationary posture is established. The `balance_robust` stage then widens friction, mass, and damping variation to cover floor surfaces from polished tile to rubber mat, battery charge variation and light payloads, actuator damping uncertainty, and communication latency through optional action delay.

Observation availability. The policy uses body-frame linear velocity, which requires a velocity estimator on hardware. Should a reliable estimator be unavailable, the velocity channel can be dropped, reducing the observation to 39 dimensions. The policy can be retrained under this reduced observation for direct deployment, at a modest cost in height-tracking precision.

Transfer plan. Planned hardware deployment will proceed in three stages: (i) system identification of actuator friction and rotor inertia to refine the simulator; (ii) policy deployment at fixed nominal height ($h^{cmd} = 0.65\,\text{m}$) with a fall-safe watchdog; (iii) progressive extension to variable-height commands following the same Phase A/B/C ordering as training.

C. Current Limitations

1) **Balance only (experimentally).** Variable-height balance is the experimentally mature component. The stand-up

and robust-balance extensions are implemented in code and configuration, but they have no training results at the time of writing. All claims about push recovery and height transitions are structural, not performance-validated.

2) **No quantitative results yet.** Training runs have not been completed. All result sections contain placeholders. Claims in this paper concern framework design, not demonstrated performance.
3) **No sim-to-real transfer (yet).** All work is simulation-only. Although the model matches a physical platform, hardware transfer has not been attempted. Sensor-noise and action-delay settings follow established sim-to-real protocols [8], [28], [29], but validation remains future work.
4) **Multi-seed training planned, not yet complete.** Final metrics will aggregate runs with seeds 42, 113, and 999. Quantitative results will report mean $\pm$ std across all three seeds.
5) **LQR at large disturbances.** The LQR is only valid near the nominal height and small lean angles. Comparisons at full range, strong pushes, and low friction are expected to favor RL, but this must be confirmed experimentally.
6) **Conservative curriculum advancement.** The within-stage height curriculum evaluates roughly every 10^7 environment steps using 256 fixed-horizon greedy episodes. Advancement additionally requires training reward, evaluation return, success rate, and fall-rate gates to pass. This reduces premature range expansion but may slow progress when a policy is close to satisfying the next curriculum level.

XI. Conclusion

This paper presented a curriculum reinforcement learning framework for variable-height balancing of a 10-actuated wheeled biped robot. The core technical contributions are: (i) a height-conditioned policy trained via PPO in GPU-accelerated MuJoCo MJX simulation with 4,096 parallel environments; (ii) a within-stage height curriculum that expands the commanded height range from near-nominal standing to deep crouching across 29 levels, gated by a greedy evaluation pass; (iii) a height-consistent natural pose reward that maps commanded height to target joint angles via piecewise-linear interpolation, preventing degenerate balancing strategies; and (iv) a classical LQR baseline sharing the same PID actuator interface, enabling controlled comparison across a suite of ten evaluation scenarios.

The framework is designed for sim-to-real transfer: PD position targets as the action representation, comprehensive domain randomization covering friction, mass, joint damping, and sensor noise, and an observation space that degrades gracefully when linear velocity is unavailable. A multi-stage macro-curriculum (`balance` $\rightarrow$ `stand_up` $\rightarrow$ `balance_robust`) provides a path toward robust recovery from falls and active height transitions.

Experimental evaluation and quantitative comparison with the LQR baseline are ongoing; results will be reported in the final version.

Future work will focus on hardware deployment via the staged transfer plan described in Section X-B, extension of the macro-curriculum to the `stand_up` and `balance_robust` stages, and integration of exteroceptive sensing for terrain-aware variable-height locomotion.

Acknowledgment

References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[2] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.

[3] "MuJoCo Documentation: MuJoCo XLA (MJX)," [Online]. Available: https://mujoco.readthedocs.io/en/latest/mjx.html. Accessed: Apr. 2026.

[4] J. Bradbury *et al.*, "JAX: composable transformations of Python+NumPy programs," 2018. [Online]. Available: http://github.com/jax-ml/jax

[5] J. Heek, A. Levskaya, A. Oliver, *et al.*, "Flax: A neural network library and ecosystem for JAX," 2024. [Online]. Available: http://github.com/google/flax

[6] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-Dimensional Continuous Control Using Generalized Advantage Estimation," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.

[7] B. P. Welford, "Note on a Method for Calculating Corrected Sums of Squares and Products," *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962.

[8] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.

[9] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum Learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2009, pp. 41–48.

[10] V. Klemm *et al.*, "Ascento: A Two-Wheeled Jumping Robot," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2019.

[11] V. Klemm, A. Morra, L. Gulich, *et al.*, "LQR-Assisted Whole-Body Control of a Wheeled Bipedal Robot with Kinematic Loops," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3745–3752, 2020.

[12] C.-F. Hsu, B.-R. Chen, and Z.-L. Lin, "Implementation and Control of a Wheeled Bipedal Robot Using a Fuzzy Logic Approach," *Actuators*, vol. 11, no. 12, p. 357, 2022.

[13] X. Li *et al.*, "Dynamic wheeled motion control of wheel-biped transformable robots," *Biomimetic Intelligence and Robotics*, 2021.

[14] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, "Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning," in *Proc. Conf. Robot Learning (CoRL)*, vol. 164, pp. 91–100, 2022.

[15] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning Quadrupedal Locomotion over Challenging Terrain," *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.

[16] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "RMA: Rapid Motor Adaptation for Legged Robots," in *Proc. Robotics: Science and Systems (RSS)*, 2021.

[17] J. Hwangbo *et al.*, "Learning Agile and Dynamic Motor Skills for Legged Robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.

[18] T. Miki *et al.*, "Learning Robust Perceptive Locomotion for Quadrupedal Robots in the Wild," *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.

[19] J. Lee *et al.*, "Learning Robust Autonomous Navigation and Locomotion for Wheeled-Legged Robots," *Science Robotics*, vol. 9, 2024.

[20] E. Vollenweider *et al.*, "Advanced Skills through Multiple Adversarial Motion Priors in Reinforcement Learning," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 5120–5126, 2023.

[21] I. Chamorro, J. Arrizabalaga, and M. Hutter, "Reinforcement Learning for Blind Stair Climbing with Legged and Wheeled-Legged Robots," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024.

[22] J. Zhu, S. Zhao, and E. Yoshida, "Whleaper: A 10-DOF Flexible Bipedal Wheeled Robot," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 11272–11277, 2024.

[23] J. Zhu and M. Hayashibe, "Reinforcement Learning Based Hierarchical Control for Path Tracking of a Wheeled Bipedal Robot with Sim-to-Real Framework," in *Proc. IEEE/SICE Int. Symp. System Integration (SII)*, 2022.

[24] S. Liu *et al.*, "DIABLO: A 6-DoF Wheeled Bipedal Robot Composed Entirely of Direct-Drive Joints," *arXiv preprint arXiv:2407.21500*, 2024.

[25] R. Portelas, C. Colas, L. Weng, K. Hofmann, and P.-Y. Oudeyer, "Automatic Curriculum Learning for Deep RL: A Short Survey," in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2020.

[26] R. Portelas, C. Colas, K. Hofmann, and P.-Y. Oudeyer, "Teacher Algorithms for Curriculum Learning of Deep RL in Continuously Parameterized Environments," in *Proc. Conf. Robot Learning (CoRL)*, vol. 100, pp. 835–853, 2020.

[27] R. Wang, J. Lehman, J. Clune, and K. O. Stanley, "Paired Open-Ended Trailblazer (POET): Endlessly Generating Increasingly Complex and Diverse Learning Environments and Their Solutions," *arXiv preprint arXiv:1901.01753*, 2019.

[28] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-Real Transfer of Robotic Control with Dynamics Randomization," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 3803–3810, 2018.

[29] J. Tan *et al.*, "Sim-to-Real: Learning Agile Locomotion for Quadruped Robots," in *Proc. Robotics: Science and Systems (RSS)*, 2018.